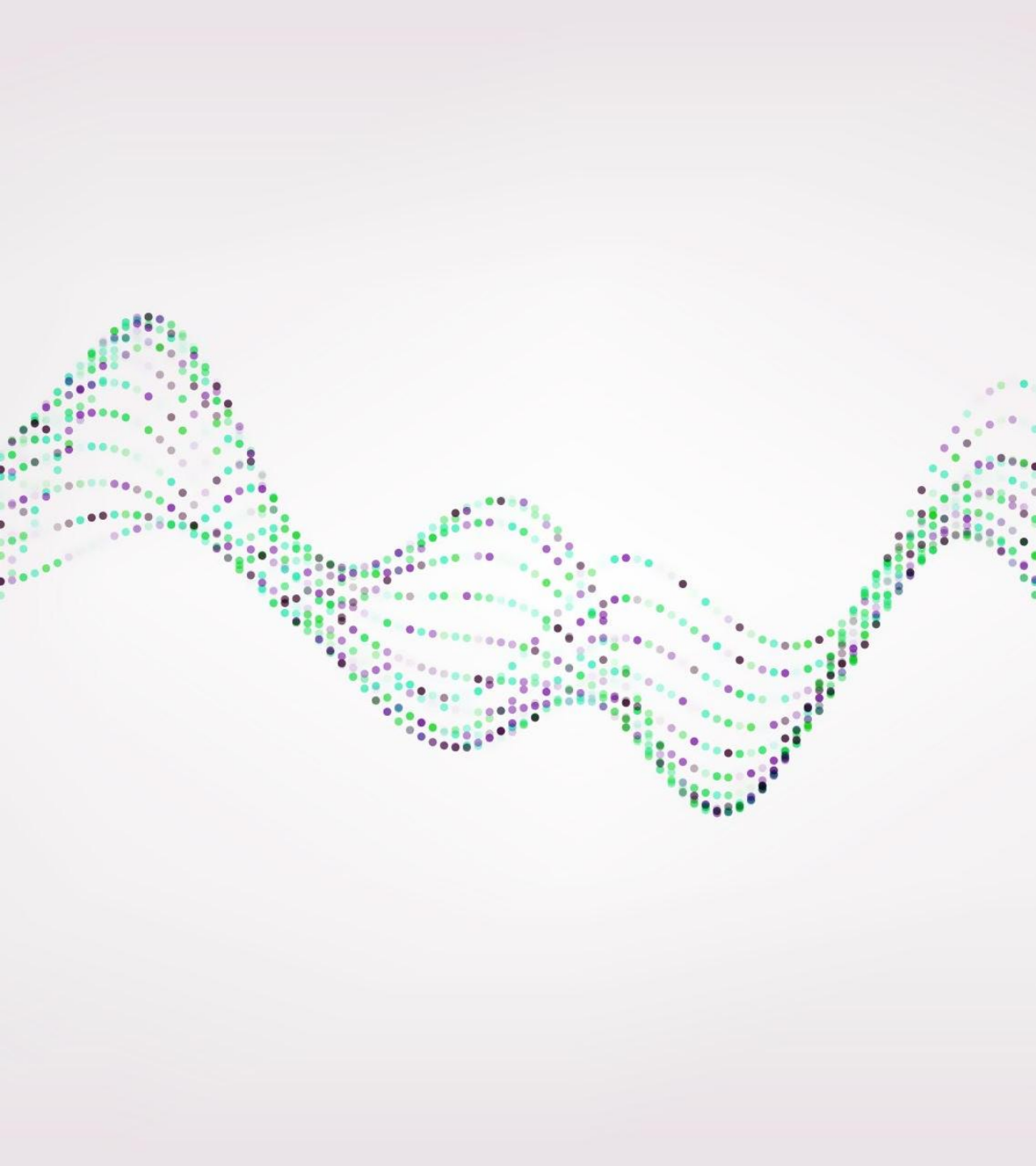




DESIGN AND IMPLEMENTATION OF AN SDR-BASED WIRELESS CHANNEL SOUNDER

Indoor Multipath Characterization using USRP B210

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WES 207 Capstone

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WHY THIS MATTERS



Wireless channels are
unpredictable indoors



Multipath causes fading, delay,
and distortion



Critical for:

Wi-Fi / 5G
performance
Radar & sensing
RF system design

PROJECT IDEA

Core

Core Concept

- Build an SDR-based channel sounder
- Use USRP B210 radios
- Measure real-world indoor propagation

Key

Key Output:

- Channel Impulse Response (CIR)
- Power Delay Profile (PDP)
- Delay Spread



SYSTEM OVERVIEW

- **High-Level Architecture**
 - **Transmitter**
 - Known probe waveform
 - **Receiver**
 - Captures IQ samples
 - **Processing**
 - Correlation $\rightarrow$ CIR $\rightarrow$ Metrics
- **Hardware & Tools**
 - USRP B210
 - Python / MATLAB
 - GNU Radio (optional)

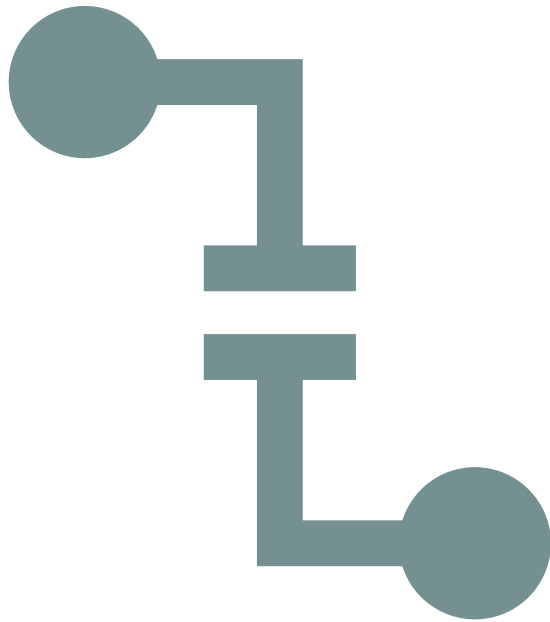
MVP DEFINITION

Minimum Viable Product (Target: Week 8)

- ✓ Establish SDR Tx/Rx link
- ✓ Transmit known waveform
- ✓ Capture IQ data
- ✓ Perform correlation
- ✓ Extract CIR
- ✓ Generate Power Delay Profile
- ✓ Test in 3 environments

Success Criteria

Clear difference between LOS vs NLOS



METRICS & OUTPUTS

- **What Will Be Measured**
 - Channel Impulse Response (CIR)
 - Power Delay Profile (PDP)
 - Delay Spread
 - Path Loss (basic)
- **Visualization**
 - Time-domain plots
 - Comparative environment plots

PROJECT TIMELINE & EXECUTION PLAN

Sprint 1 — System Setup (Weeks 1–2)

- Setup USRP + environment
- Verify Tx/Rx link

Sprint 2 — Signal Development (Weeks 3–4)

- Implement waveform transmission
- Capture IQ data

Sprint 3 — Signal Processing (Weeks 5–6)

- Correlation + CIR extraction
- Generate PDP

Sprint 4 — Testing & Analysis (Weeks 7–8)

- Conduct LOS/NLOS experiments
- Plot + analyze results

EXTENSIONS

Signal Processing

Multi-bandwidth comparison

Frequency-selective analysis

Advanced Features

FFT-based channelizer

Coherence bandwidth estimation

System Upgrades

Real-time visualization

MIMO channel sounding

Doppler analysis

Hardware Acceleration

FPGA-based processing

WHY THIS PROJECT (PERSONAL + IMPACT)

Relevance

- Aligns with:
 - Atmospheric Propagation (NIWC Pacific)
 - Wireless system design
 - Radar & sensing applications

Personal Motivation

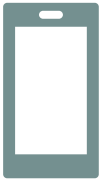
- Background in weather + propagation
- Current work in RF/propagation
- Future research alignment

Impact

- Bridges:
 - Theory
 - Signal processing
 - Real-world RF systems



KEY TAKEAWAYS



**Designing and
implementing a real SDR-
based channel sounder**



**Focus on measurable
propagation metrics**



**MVP achievable within 8
weeks**



**Strong alignment with
real-world RF systems**

FEEDBACK & QUESTIONS

1. Technical Approach

- Is **correlation-based channel sounding** sufficient for MVP?
- Any recommended **probe waveforms** (PN sequence, chirp, OFDM)?

2. Scope & Feasibility

- Is the **MVP appropriately scoped** for this timeline?
- Any **risks** I should prioritize early?

3. Extensions & Direction

- Which extensions add the most value?
 - **Channelizer**
 - **MIMO**
 - **Real-time processing**

